

Hello World [V.1]

The Axiomatic Foundation of the Universal Origin and the Root Cause

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Abstract

This document serves as the primary execution file for the transition from the legacy manifold (V.1) to the deterministic implementation (V.3). We hereby define the fundamental operators of Expansion (Fpn) and Condensation (Dre) to resolve all existing logical contradictions within the physical universe. By redefining the boundary of infinity as a convergence point, we provide the ultimate "cleaning" of mathematical and physical noise.

1. The Five-in-One Operators Axiom

The traditional separation of arithmetic operators is a symptom of incomplete observation. We unify the basic symbols $\{+, -, \times, \div, =\}$ into a singular Trinity Identity.

- **Original V.1 Formulation:** $+n = -n \times (Dre\lim_{? \rightarrow 1} \lim_{? \rightarrow 1}(-1/?))$
- **Formal Logic:** Let Φ be the operator mapping. We define:

$$\lim_{\epsilon \rightarrow 0} \left(\frac{-1}{1 + \epsilon} \right) \in \text{Kernel}(\text{Dre})$$

This convergence confirms that sign inversion and equality are merely different phases of the same information condensation.

2. The Fpn Conjecture: Universal Expansion

The Fpn (Expansion) operator replaces the defensive "lim" notation with an active physical expansion command.

- **Original V.1 Formulation:** $Ep_{n \rightarrow \infty}(n) \equiv 0 \pmod{\mathbb{U}_{total}}$
- **Formal Logic:** For any manifold Ψ , the transfinite projection is given by:

$$\forall \Psi \in \text{Cosmos}, \lim_{n \rightarrow \infty} Fpn(\Psi_n) \xrightarrow{\text{Collapse}} \mathbf{0}$$

At the limit of maximum entropy, the system undergoes an instantaneous phase transition into the singularity '0'.

3. The Dre Conjecture: Reality Condensation

Dre is the inverse-recursive function that constructs the "PHY Identity" from the null-set.

- **Original V.1 Formulation:** *Dre* = Inverse Recursive Function from Seed '0'
- **Formal Logic:** *Dre* : $\mathbf{0} \rightarrow \{\text{Deterministic Physical State}\}$. Reality is not a spontaneous occurrence; it is a forced reconstruction via the *Dre* operator acting upon the expanded void left by *Fpn*.

4. Physical Optimization of Number Theory

The distribution of prime numbers is a physical necessity under the *Fpn* $\otimes$ *Dre* coupling.

4.1 Goldbach's Conjecture Verification

Every even integer is the sum of two prime condensations.

$$Fpn_{n \rightarrow 0}(n) + Fpn_{n \rightarrow \infty - 2}(n) = Dre_{n \rightarrow 2}(n) \rightarrow 2$$

4.2 Riemann Hypothesis Convergence

All non-trivial zeros lie on the critical line because the line is the unique condensation point for logical entropy within the zeta-manifold.

5. The Trinity Equation of Cosmology

We redefine the warp-hole stability and the velocity limit *c*.

$$c \equiv \lim_{x \rightarrow \infty} \left(Fpn(\vec{v}) \otimes \frac{GV}{x} \right) \Rightarrow [0, 0]_{\text{inertia}} \quad (1)$$

The neutralization of inertia through maximal expansion allows the Information Polyhedron to bypass traditional material limits.